

UK RETAIL COMMERCIAL INTELLIGENCE PLATFORM

Project Conclusion & Solution Summary

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1. EXECUTIVE SUMMARY

This project set out to build a full-stack commercial analytics platform on Databricks that replicates the real-world responsibilities of a Data Analyst in a UK retail environment. The platform ingests, cleans, models, and surfaces commercial intelligence across the complete product lifecycle — from raw point-of-sale transactions through to board-ready insights and auto-generated business recommendations.

The business context simulated a design-led UK nursery brand distributing through eight major retail partners including Boots, John Lewis, Asda, and Amazon UK, alongside a growing DTC ecommerce channel. The dataset covered over 958,000 UK transactions across two years.

£17.9M

Gross Revenue
analysed

£15.0M

Net Revenue
(post-deductions)

£8.6M

Contribution
Margin delivered

45%

Average Margin %

£2.85M

Margin Leakage
identified

2. THE PROBLEM

UK retail brands operating across multiple channels and retailer partners face a persistent set of commercial challenges that raw data systems fail to address:

2.1 Margin Opacity

Gross revenue figures mask the true commercial performance. Rebates, settlement discounts, scan allowances, and cost of goods sold collectively erode margin — but these deductions are rarely modelled together in a single view. Without a clear Gross to Net to Contribution Margin waterfall, brands cannot identify which retailers, categories, or SKUs are genuinely profitable.

2.2 Promotional Waste

Trade investment is one of the largest cost lines in a retail brand's P&L, yet most promotional analysis stops at reporting total sales during a promotion. The critical question — was this promotion incremental, or did customers simply buy what they would have bought anyway? — goes unanswered. Value-destroying promotions continue to be funded because the ROI evidence is never surfaced.

2.3 Reactive Rather Than Predictive Trading

Without a structured forecasting capability, commercial teams respond to trading performance rather than anticipating it. Seasonal demand patterns, slow-moving stock, and channel performance shifts are identified too late to act on effectively. Sell-through deterioration becomes a write-off risk before anyone flags it.

2.4 Pricing Without Evidence

Pricing decisions are often made based on intuition, competitor observation, or historical precedent rather than quantified demand analysis. Without understanding which categories are price-elastic versus inelastic, brands risk leaving margin on the table in inelastic categories or destroying volume in elastic ones.

2.5 Fragmented Intelligence

Market share, competitor pricing, regional whitespace, and seasonal risk each sit in separate tools or manual spreadsheets. There is no unified view that connects external market signals to internal commercial performance and surfaces actionable recommendations in a single place.

3. THE SOLUTION

The UK Retail Commercial Intelligence Platform addresses each of these problems through a structured, end-to-end analytics build on Databricks. The solution is organised into six analytical modules, each targeting a specific commercial challenge, all feeding into a unified AI/BI dashboard.

3.1 Commercial P&L Engine (Module 1)

Solution to Problem 2.1 — Margin Opacity

A complete Gross to Net Revenue to Contribution Margin waterfall was built at retailer, channel, region, and SKU level. The model applies actual rebate rates, settlement discount percentages, scan allowances, and cost of goods sold from the synthetic ERP product master to every transaction.

- Identified £2.85M in margin leakage across the portfolio
- Produced retailer-level P&L with RAG status flags (Green, Amber, Red)
- Ranked 4,906 SKUs by contribution margin — surfacing HERO, SOLID, WATCH, and AT RISK classifications
- Built a 7-step margin waterfall bridge table for Power BI visualisation

3.2 Trade Spend & Promotional ROI Engine (Module 2)

Solution to Problem 2.2 — Promotional Waste

A promotional incrementality model was built using a rolling 4-week pre-promotion baseline to separate genuine incremental volume from substitution. Every promotion was assigned a net ROI after trade spend costs.

- Calculated promotional ROI for all 12 promotion events across 8 retailers
- Flagged VALUE DESTROYING promotions where trade spend exceeded incremental margin
- Built a trade spend efficiency heatmap — spend per incremental unit by retailer
- Ranked all promotions best to worst ROI with STRONG POSITIVE / VALUE DESTROYING flags

3.3 Forecasting & Sell-Through Engine (Module 3)

Solution to Problem 2.3 — Reactive Trading

A Prophet time-series forecasting model was trained per channel with Base, Upside (+15%), and Downside (-20%) scenarios, all tracked and versioned in MLflow. A sell-through rate engine monitors weekly SKU performance against a rolling 12-week availability benchmark.

- Prophet models trained per channel with yearly and weekly seasonality
- Three-scenario forecast output — Base, Upside, Downside — for trading review cadence
- Sell-through rate calculated per SKU per week with FAST MOVER / AT RISK flags
- Seasonal risk index built per category and quarter — TROUGH and PEAK periods identified
- All models logged in MLflow with MAPE and bias metrics tracked

3.4 Pricing Elasticity & Market Intelligence (Module 4)

Solution to Problem 2.4 — Pricing Without Evidence

Price elasticity of demand was calculated for all 8 product categories using OLS regression on log-transformed price and volume data. Competitor pricing was indexed against the brand's average selling prices across five simulated UK nursery competitors.

- Elasticity coefficients calculated per category: HIGHLY ELASTIC, ELASTIC, MILDLY ELASTIC, INELASTIC
- Inelastic categories flagged as price increase opportunities with quantified impact estimates
- Competitor pricing index built — HIGH / MODERATE / LOW risk flags per category
- UK regional whitespace analysis: revenue share vs population weight gap per region
- London identified as MAJOR WHITESPACE — 0% revenue share vs 13.4% population weight

3.5 Recommendations Engine (Module 6)

Solution to Problem 2.5 — Fragmented Intelligence

A structured business rules engine queries all 18 Gold Delta tables and auto-generates commercial recommendations with evidence, recommended action, and commercial impact for each flagged issue. A weekly trading briefing is auto-generated in markdown format, replicating the output of a commercial review pack.

- 10 business rules applied across all commercial dimensions
- 34 HIGH priority recommendations generated
- 6 MEDIUM priority recommendations generated
- Every recommendation includes: Issue, Evidence (metric), Recommended Action, Commercial Impact, Priority
- Weekly trading briefing auto-generated from Gold KPIs — ready for stakeholder distribution

4. KEY COMMERCIAL FINDINGS

Finding	Evidence	Recommended Action	Priority
£2.85M margin leakage across portfolio	Rebates + settlement + scan allowances vs gross revenue	Negotiate improved funding terms with high-leakage retailers	HIGH
VALUE DESTROYING promotions identified	Negative ROI on multiple promotion events	Redesign or cancel — reduce discount depth or shift mechanics	HIGH
AT RISK SKUs with sub-30% margin	gold_sku_profitability — margin rank and flag	Review cost structure or delist — range rationalisation required	HIGH
DTC margin 26.8% vs Retail 63.3%	Channel P&L comparison in gold_retailer_pl	Investigate DTC cost structure — review pricing and fulfilment costs	HIGH
London = MAJOR WHITESPACE (0% revenue)	Revenue share 0% vs population weight 13.4%	Engage London-strong retail partners and increase distribution	MEDIUM
Inelastic categories identified	OLS elasticity coefficients below -0.5 threshold	Test 5-8% RRP increase — minimal volume loss expected	HIGH
Seasonal troughs in multiple categories	Seasonal index below 0.7 in 2+ quarter periods	Plan promo investment and NPD timing around trough periods	MEDIUM
Poor forecast accuracy in some channels	MAPE above 30% threshold in gold_forecast_accuracy	Incorporate promotional calendar into forecast model inputs	MEDIUM
Competitor price undercuts in key categories	Price gap analysis in gold_competitor_pricing	Selective price match on top 3 volume SKUs per category	HIGH
Slow-moving SKUs — 4+ consecutive weeks	Sell-through below 40% threshold in gold_sell_through	Clearance markdown or promotional clearance event required	HIGH

5. CONCLUSION

This project demonstrates that a commercial analyst with access to the right data infrastructure can move far beyond descriptive reporting and into genuine decision support. The platform built here does not simply report what happened — it interrogates performance, quantifies the commercial impact of every decision, and recommends specific actions with evidence.

Three outcomes stand out as commercially significant from this analysis:

- **The margin leakage analysis revealed that £2.85M — or approximately 15.9% of gross revenue — is being consumed by trade funding deductions before a single unit of cost of goods is deducted. This is structural margin leakage that could be partially recovered through renegotiation of retailer funding terms, particularly with the highest-volume partners where even a 1% improvement in rebate rate drives meaningful P&L recovery.**
- **The promotional ROI analysis exposed a pattern of trade investment that is not generating returns. Several promotions delivered negative net incremental margin — meaning the brand spent more on the promotion than it earned from the additional sales it drove. These promotions persist because the incrementality calculation is rarely done. This platform makes it automatic.**
- **The regional whitespace analysis reveals that London — the UK's largest consumer market — accounts for 0% of the brand's revenue share despite representing 13.4% of the national population. This is not a data anomaly. It is a structural distribution gap that represents the single largest growth opportunity in the portfolio.**

From a technical standpoint, the project validates Databricks as a production-grade platform for commercial analytics — not just data engineering. The combination of PySpark for transformation, Delta Lake for reliable storage, Prophet and scikit-learn for ML, MLflow for model governance, and the AI/BI dashboard for stakeholder delivery creates a complete, scalable, and maintainable commercial intelligence stack.

The platform is designed to be extended. New retailer data, updated promotional calendars, or revised cost structures can be ingested into the Bronze layer and the entire Gold layer will refresh automatically. The recommendations engine will surface updated priorities without manual intervention.

6. LESSONS LEARNED

Area	Lesson Learned
Data Quality	The UCI Online Retail II dataset required significant cleaning — cancellation invoices, negative quantities, zero-price rows, and null customer IDs all needed handling before any commercial analysis could run. Real retail POS data is always messier than expected.

ERP Integration	Without a product master (cost prices, category, supplier), contribution margin cannot be calculated. Building the synthetic ERP layer was essential and mirrors the real-world dependency analysts have on ERP data.
Promotional Incrementality	Calculating a reliable baseline requires enough historical data before the promotion window. Promotions with insufficient pre-period data produced unstable baselines — a real constraint in live commercial environments.
MLflow on Serverless	Prophet installation on Databricks Serverless required a kernel restart after pip install. This is a known limitation of the Serverless compute environment and needs to be handled carefully in production pipelines.
Delta Lake Column Names	Delta Lake does not allow special characters in column names. The ONS retail datasets contained square brackets and commas in column headers — all required cleaning with regex before writing to Delta tables.
Dashboard Filter Architecture	Databricks AI/BI filters only work across datasets when the filter column is explicitly included in every dataset SQL query. This requires deliberate dataset design upfront — adding channel, category, region, and month to every GROUP BY.
Commercial Storytelling	The most important lesson is that technical accuracy is necessary but not sufficient. Every chart, every table, and every recommendation needs to tell a commercial story that a senior stakeholder can act on immediately.

7. WHAT IS NEXT

Immediate Extensions

- Push all 6 notebooks to GitHub with a full README — make the project publicly discoverable
- Publish the LinkedIn carousel (3 slides) to attract UK retail analytics recruiters
- Add the project to CV under Data Projects with key metrics and tech stack

Technical Enhancements

- Build a Databricks Workflow to schedule the full pipeline — Bronze through Gold — on a weekly cadence
- Add data quality checks using Great Expectations or Databricks built-in expectations
- Integrate real ONS baby and nursery category data for more accurate market benchmarking
- Build a proper Star Schema in the Gold layer with fact and dimension tables
- Add Unity Catalog data lineage tracking across all tables

Commercial Extensions

- Model customer lifetime value (CLV) per retailer segment
- Add a demand sensing layer using Google Trends data as a leading indicator for the Prophet forecast
- Build a net revenue management (NRM) simulator — model the P&L impact of changing funding rates

- Add competitor NPI (new product introduction) tracking to the market intelligence module